

Surface Area of Prisms

Lesson 10-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Paint every outside surface of a box — the total painted area is the surface area

Surface area

The total area of all the flat sides of a solid.

A box with 6 flat rectangle sides: top, bottom, front, back, left, right

Rectangular prism

A solid box shape with six flat rectangle sides.

Shaped like a tent or a wedge of cheese — 2 triangle ends + 3 rectangle sides

Triangular prism

A solid with two triangle ends and three flat rectangle sides.

A triangular prism has 5 faces: 2 triangles + 3 rectangles

Face

One flat side of a solid shape.

Unfold a triangular prism flat: you see 2 triangles and 3 rectangles side by side

Net

A flat shape that folds up into a solid.

On a triangular prism, the 3 rectangles that wrap around the sides are lateral faces

Lateral face

The side faces of a solid, not the top or bottom.

Key Ideas & Notes

- Your class is painting the outside of different time capsule shapes to protect them from weather.
- Some capsules are rectangular prisms and others are triangular prisms (wedge-shaped).
- To buy the right amount of paint, you need to calculate the surface area of each shape!
- Calculate the surface area of each time capsule shape. For rectangular prisms, use $SA = 2lw + 2lh + 2wh$. For triangular prisms, find the area of each face and add them up.

Think About It

- How many faces does a rectangular prism have? A triangular prism?
- What shapes make up the faces of a triangular prism?
- How is finding surface area different from finding volume?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Your class is painting two capsule shapes: a rectangular prism and a triangular prism. How is finding the surface area of each shape the same, and how is it different?

Level 1 support

Start here: For both, you add the areas of all the faces, but the shapes and number of faces differ.

Both surface areas are found by adding the areas of all the ____.

Ambas áreas de superficie se hallan sumando las áreas de todas las ____.

A rectangular prism has ____ faces, but a triangular prism has ____ faces.

Un prisma rectangular tiene ____ caras, pero un prisma triangular tiene ____ caras.

Word bank: surface area rectangular prism triangular prism face lateral face

Listen for: Students explain both methods add up all face areas, but a rectangular prism has 6 rectangular faces while a triangular prism has 5 faces (2 triangles + 3 rectangles).

Level 2

Why does surface area use square units while volume uses cubic units? Justify with what each measurement counts.

- Surface area uses square units because it measures ____.
- Volume uses cubic units because it measures ____.

EXPLORE

For the triangular prism (base 6, height 4, length 10, slant sides 5), how did you account for the triangular bases AND the rectangular lateral faces?

Level 1 support

Start here: Add the two triangular bases to the three rectangular lateral faces.

The two triangular bases together are $2 \times (\frac{1}{2} \times 6 \times 4) = \underline{\hspace{1cm}}$.

Las dos bases triangulares juntas son $2 \times (\frac{1}{2} \times 6 \times 4) = \underline{\hspace{1cm}}$.

The three lateral faces add to $\underline{\hspace{1cm}}$, so the total surface area is $\underline{\hspace{1cm}}$.

Las tres caras laterales suman $\underline{\hspace{1cm}}$, así que el área total de superficie es $\underline{\hspace{1cm}}$.

Word bank: (surface area) (triangular prism) (lateral face) (face) (net)

Listen for: Students compute the two triangle bases (24 in^2) and the three rectangles ($60 + 50 + 50 = 160 \text{ in}^2$), totaling $SA = 184 \text{ in}^2$, and distinguish bases from lateral faces.

Level 2

Compare the rectangular capsule ($SA = 248 \text{ in}^2$) with the similar triangular capsule ($SA = 184 \text{ in}^2$). Why does the rectangular one need more paint?

- The rectangular prism needs more paint because it has $\underline{\hspace{1cm}}$ faces and $\underline{\hspace{1cm}}$.
- Even at a similar size, the triangular prism uses less because $\underline{\hspace{1cm}}$.

CONNECT

A carpenter stains a chest that is $3\text{ ft} \times 2\text{ ft} \times 2\text{ ft}$, and one can covers 50 ft^2 . Talk through how to find the surface area and how many cans are needed.

Level 1 support

Start here: Find the surface area with $SA = 2lw + 2lh + 2wh$, then divide by 50 and round up.

The chest's surface area is ____ ft^2 because $SA = 2(3 \times 2) + 2(3 \times 2) + 2(2 \times 2)$.

El área de superficie del cofre es ____ ft^2 porque $SA = 2(3 \times 2) + 2(3 \times 2) + 2(2 \times 2)$.

The carpenter needs ____ can(s) because ____ $\div 50 =$ ____.

El carpintero necesita ____ lata(s) porque ____ $\div 50 =$ ____.

Word bank: (surface area) (rectangular prism) (face) (lateral face) (net)

Listen for: Students compute $SA = 12 + 12 + 8 = 32\text{ ft}^2$, then $32 \div 50 < 1$, and reason that 1 can is enough since the surface area is under 50 ft^2 .

Level 2

If the chest had a lid that does not get stained, how would that change the surface area you calculate? Explain.

- Leaving the lid unstained changes the surface area by ____ because ____.
- I would subtract ____ from the total because ____.

REFLECT

A classmate wrote $SA = 2(12 \times 7) + 2(12 \times 5) + (7 \times 5) = 323 \text{ cm}^2$ for a $12 \times 7 \times 5$ prism. How do you find and fix their mistake?

Level 1 support

Start here: They forgot to multiply the last face pair by 2; the correct SA is 358 cm^2 .

The mistake is that the last pair (7×5) was not multiplied by ____.

El error es que el último par (7×5) no se multiplicó por ____.

The correct surface area is ____ cm^2 because $2(7 \times 5) =$ ____, not ____.

El área de superficie correcta es ____ cm^2 porque $2(7 \times 5) =$ ____, no ____.

Word bank: (surface area) (rectangular prism) (face) (lateral face) (net)

Listen for: Students notice the missing '2' on the last pair, compute $2(7 \times 5) = 70$, and correct the total to $168 + 120 + 70 = 358 \text{ cm}^2$.

Level 2

What general rule about prism faces does this error break, and how could the net help a student avoid it?

- The error breaks the rule that opposite faces ____.
- Drawing the net would help because ____.

Guided Examples

Example 1

What is the surface area of a rectangular prism with $l = 5$ cm, $w = 4$ cm, $h = 3$ cm?

Solution: $SA = 2(5 \times 4) + 2(5 \times 3) + 2(4 \times 3) = 40 + 30 + 24 = 94 \text{ cm}^2$.

Answer: A. 94 cm^2

Example 2

How many faces does a triangular prism have?

Solution: A triangular prism has 5 faces: 2 triangular bases and 3 rectangular lateral faces.

Answer: A. 5

Example 3

A triangular prism has two triangular bases each with area 12 cm^2 and three rectangular faces with areas 30 cm^2 , 40 cm^2 , and 40 cm^2 . What is the total surface area?

Solution: $SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134 \text{ cm}^2$.

Answer: A. 134 cm^2

Write About the Math

The Writing Revolution

I can explain my work using the words surface area, rectangular prism, triangular prism, and face.

1. Kernel Sentence subject + verb

Model: Surface area is the total area of all the flat sides of a solid.

Área de superficie es el área total de todos los lados planos de un sólido.

Write a kernel sentence about surface area. Use a subject and a verb.

Escribe una oración base sobre área de superficie. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Surface area matters in math

Área de superficie importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Surface area matters in math because ____.

Área de superficie importa en matemáticas porque ____.

but
pero

Surface area matters in math, but ____.

Área de superficie importa en matemáticas, pero ____.

so
entonces

Surface area matters in math, so ____.

Área de superficie importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about surface area.
Di un hecho verdadero sobre surface area.

Surface area ____.

Question *Pregunta*

Ask a question about surface area.
Haz una pregunta sobre surface area.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about surface area.
Muestra entusiasmo sobre surface area.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with surface area.
Dile a un compañero qué hacer con surface area.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

Both surface areas are found by adding the areas of all the ____.

Ambas áreas de superficie se hallan sumando las áreas de todas las ____.

A rectangular prism has ____ faces, but a triangular prism has ____ faces.

Un prisma rectangular tiene ____ caras, pero un prisma triangular tiene ____ caras.

The two triangular bases together are $2 \times (\frac{1}{2} \times 6 \times 4) =$ ____.

Las dos bases triangulares juntas son $2 \times (\frac{1}{2} \times 6 \times 4) =$ ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A triangular prism has two triangular bases each with area 12 cm^2 and three rectangular faces with areas 30 cm^2 , 40 cm^2 , and 40 cm^2 . What is the total surface area?

A. 134 cm^2

B. 110 cm^2

C. 122 cm^2

D. 134 cm^3

Show your work:

2. A triangular prism has triangular bases with base 6 in and height 4 in. The prism is 10 in long and the two slant sides of the triangle are each 5 in. What is the surface area?

A. 184 in^2

B. 160 in^2

C. 124 in^2

D. 184 in^3

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A company ships products in two container options: a rectangular prism ($8 \times 6 \times 4$ in) or a triangular prism (triangle base 8 in, triangle height 6 in, prism length 8 in, slant sides 6.3 in each). Both must be wrapped in protective film. Which shape uses LESS film? Show your work.

Sentence starter: Rectangular prism SA = $2(\text{ }) + 2(\text{ }) + 2(\text{ }) = \text{ } \text{ in}^2$. Triangular prism SA = $2(\text{ }) + \text{ } + \text{ } + \text{ } = \text{ } \text{ in}^2$. The uses less film because .

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 8$ ft, $w = 6$ ft, $h = 3$ ft. What is the surface area?

- A. 180 ft^2
- B. 144 ft^2
- C. 180 ft^3
- D. 90 ft^2

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 134 cm^2 — $SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134 \text{ cm}^2$.
2. **Try It 2:** A. 184 in^2 — *Triangular bases:* $2 \times (\frac{1}{2} \times 6 \times 4) = 24 \text{ in}^2$. *Three rectangular faces:* $6(10) + 5(10) + 5(10) = 60 + 50 + 50 = 160 \text{ in}^2$. $SA = 24 + 160 = 184 \text{ in}^2$.
3. **Exit Ticket:** A. 180 ft^2 — $SA = 2(8 \times 6) + 2(8 \times 3) + 2(6 \times 3) = 96 + 48 + 36 = 180 \text{ ft}^2$. *Surface area uses square units (ft^2).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Surface area is the total area of all the flat sides of a solid.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).